UTHEALTH HOUSTON AND MEMORIAL HERMANN HIGH RISK LIVER TRANSPLANT PROTOCOL FOR INTRAHEPATIC CHOLANGIOCARCINOMA V7

This pathway addresses advanced liver-confined iCCA outside the small solitary cirrhotic category. It requires treatment-response selection, repeated resection assessment and a separate allocation/donor justification. Pure iCCA and mixed HCC–iCCA are not pooled as one efficacy population.

Version 7 | Revised 2026-10-24 06:55 CDT | Institutional approval pending

POLICY AND ALLOCATION STATUS

Analyst A structured critical review through 24 October 2026. All 27 B decisions remain OPEN. Primary evidence, selected-recipient outcomes, pathway-level outcomes, guideline wording and local safeguards are distinguished. This is not a PRISMA-complete systematic review, meta-analysis, fresh external-analyst consensus or institutional approval.

Policy posting checked 24 October 2026: OPTN effective 1 October 2026; oncology guidance December 2025. Clinical suitability, allocation access, donor protection and research authorization remain separate. Recheck before any application. [1] [2]

Use the adjacent study tables before interpreting a proposed cutoff. Abstract-only extraction and missing full PDFs are labeled; missing outcome estimates are not inferred from graphs or from one minus disease-free survival. Linked full text is in this document's own PDF depot when available.

1 PURPOSE AND SCOPE

A biopsy-proven, unresectable solitary iCCA or mixed HCC–iCCA no larger than 3 cm in cirrhosis, with the specified treatment and stability requirements, belongs in the guidance-aligned nonstandard-exception assessment. Larger, multifocal or noncirrhotic disease must not inherit that guidance by analogy. [2]

The advanced-disease pathway is a proposed prospective treatment-selection and transplant study, not a standing approval to transplant. It has no validated universal 8 cm, three-lesion, CA19-9 or ctDNA eligibility boundary.

2 SCIENTIFIC RATIONALE AND PATIENT COUNSELING

Very-early incidental explant series support further assessment of small tumors, not an advanced-disease population. In the expanded Houston experience, 65 were referred, 37 listed, five converted to resection, 32 remained in the analyzed listing cohort and 18 received LT. Reported survival was anchored to listing, including the eventual-LT subgroup. Do not label those figures post-LT survival or pool the earlier overlapping Lunsford series as independent evidence. [3] [4] [5]

SYSTARE included 13 entrants, nine dropouts and four recipients. A seven-recipient LDLT report estimated five-year OS of 85.7% with a wide 95% CI of 33.4–97.9%, two recurrences and one death. These data show selection and feasibility, not randomized benefit. The latter publisher full text was reviewed but its local PDF remains pending. [6] [7]

A non-LT eligibility cohort found 39 retrospectively eligible patients among 1407. Median OS was 39 months from diagnosis; three-/five-year survival from the six-month eligibility landmark was 46%/26%. Many received HAI. It is a relevant warning against all-comer controls, not a causal LT comparison or a current-treatment benchmark with identical eligibility. [8]

3 PATIENT SELECTION CRITERIA

3.1 ESTABLISH DIAGNOSIS AND THE EXACT RISK STRATUM

- Obtain expert pathology review of a planned core biopsy, including differentiation and any mixed HCC–iCCA component. Biopsy is part of the iCCA diagnostic/guidance pathway; do not import the pCCA primary-transperitoneal-biopsy exclusion as a blanket prohibition on iCCA biopsy. Plan the route with radiology and the surgical team. [2] [9]
- Map pretreatment lesion number, largest and aggregate burden, segment/lobe distribution, vascular anatomy, regional/distant nodes and all extrahepatic sites. Record cirrhosis and the reason complete clearance by resection/ablation is not feasible with an adequate functional remnant.
- Require documented medical fitness, functional status, liver/renal reserve and ability to tolerate the proposed therapy. Age is a risk factor for multidisciplinary assessment, not a new national cutoff invented by this document. [10]

ICCA-B1 — DO WE PRESERVE SEPARATE SMALL-CIRRHOTIC AND ADVANCED-DISEASE PATHWAYS?

QUESTION AND COMPARISON

For advanced liver-confined iCCA, how far may tumor burden, differentiation and histology be expanded beyond the small-cirrhotic pathway without treating experimental selection as guideline eligibility?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Sapisochin 2016 · very early iCCA [3] International retrospective explant-defined pure iCCA; 48 recipients, 15 very-early solitary ≤ 2 cm, 33 larger/multifocal. Primary source | Full-text PDF Locator: Hepatology 2016; 64: 1178–1188; Methods and very-early/advanced survival analyses; full PDF. Review depth: Local full PDF available/reviewed	**Reported result:** Five-year post-LT OS 65% very-early versus 45% advanced; 5-year recurrence 18% versus 61%. **Do not infer:** Small retrospective pathological strata, not validation of ≤ 3 cm guidance or response-selected large disease. Preoperative sampling/recognition differs from known iCCA.

Study • population • source	Results • interpretation boundary
De Martin 2020 [11] Three-center retrospective cirrhosis cohort; 49 LT versus 26 resection, iCCA/mixed ≤ 5 cm. Primary source | TMC: retrieve PDF Locator: Liver Transpl 2020; 26: 785–798; Methods, OS/RFS and pathological-risk analyses. Review depth: Publisher full HTML methods, pathology, survival and subgroup tables reviewed; PDF pending.	**Reported result:** From surgery, 5-year OS 67% LT versus 40% resection (p=.17), RFS 75% vs 36%(p=.004), recurrence 18% vs 46%. Differentiation independently predicted recurrence (HR 3.74). **Do not infer:** Not proof to prospectively transplant known 5-cm pure/mixed disease; heterogeneous histology, small numbers, per-protocol selection.
Dageforde 2021 [12] 12-center retrospective; 208 mixed tumors, 99 LT and 109 resection; 67 LT within Milan, 14 beyond Milan/within UCSF, 18 beyond UCSF. Primary source | Full-text PDF Locator: J Am Coll Surg 2021; 232: 361–371; pp 363–367, Tables 1–3 and tumor-burden subgroup results; local PDF. Review depth: Complete local publisher PDF inspected, especially pp363–367, Tables 1–3 and tumor-burden subgroups.	**Reported result:** Within Milan 5-year post-LT OS 70.1%, recurrence 23.1%; HCC comparators 73.4%/11.5%. Beyond UCSF mixed 5-year OS 15.8%. Within Milan LT versus resection 5-year DFS 70.3% vs 33.6%. **Do not infer:** Benefit is concentrated in small selected disease; does not validate advanced known mixed inclusion. Explant classification and preoperative misdiagnosis constrain applicability.
McMillan 2022 · Houston update [4] 65 referred, 37 listed; five converted to resection, 32 LT-pathway analysis, 18 LT. Primary source | TMC: retrieve PDF Locator: AJT 2022; 22: 823–832; Methods Statistics, Tables 1–2 and §3.5. Review depth: Publisher full HTML methods, patient flow, tables and outcomes inspected; complete PDF not found locally.	**Reported result:** Five-year OS 49% all 32 and 57% recipient subgroup, each FROM LISTING; 7/18 recurrences. **Do not infer:** Overlap with Lunsford; selection and immortal-time bias. Inclusion of G3/multifocal disease does not establish their safety, but contradicts claiming such patients were universally excluded by evidence.

Study • population • source	Results • interpretation boundary
SYS-TARE 2024 [6] Prospective institutional SYS-TARE pathway; 13 entered, 4 LT, 9 dropout. Primary source | Full-text PDF Locator: Transpl Int 2024; 37: 13 641; Eligibility, Figure 1 and Results; full PDF. Review depth: Full text reviewed	**Reported result:** Four recipients alive without recurrence at 73, 40, 12, 8 months; dropouts: three unrecovered intrahepatic progression, four with extrahepatic disease, two occult nodes. **Do not infer:** Only 4 recipients; no validated maximum diameter or comparative proof of satellites being equivalent to a solitary tumor.
Yaqub 2025 [13] Norwegian prospective early report; 7 referred, 6 enrolled, 5 LT. Primary source | TMC: retrieve PDF Locator: BJS 2025; 112: znaf054; main text, Table 1 and Figure 1; complete indexed letter. Review depth: Complete indexed primary research-letter text and Table 1/Figure 1 read; supplementary eligibility file not independently reviewed; PDF download failed.	**Reported result:** All five alive at 10–55 months; 2 recurrences at 12/13 months, both patients with >20 lesions. One HAT required re-LT; one occult 2-mm positive node. **Do not infer:** Research letter, very small denominator; recurrence pattern argues caution with diffuse multifocal disease, not a validated lesion-count cutoff.
Giguet 2026 [14] Rennes retrospective SIRT pathway; 147 treated, 44 younger controlled cases, 19 resections, eight listed including two external referrals, six LT. Primary source | TMC: retrieve PDF Locator: JHEP Rep 2026; 8: 101 830; Eligibility, cohort flow, Tables 1–3. Review depth: Indexed complete publisher full text, Tables 1–3, cohort flow and late complications reviewed; downloadable PDF blocked. Six transplanted, not eight.	**Reported result:** Post-LT 5-year OS 100%, PFS 44.4%; 3/6 recurrences. **Do not infer:** Only six recipients over 15 years; eight refers to listing, not completed LT. No size cap validated.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

Small incidental iCCA/mixed disease cannot simply be pooled with known advanced pure iCCA. Large unifocal tumors can have durable post-LT survival after selection, but recurrence remains common and the screened-to-LT yield is low. Multifocality and poor differentiation are adverse signals; their precise exclusion boundaries remain unvalidated. Known advanced mixed tumors have particularly weak prospective support. [3] [11] [12] [4] [6] [13] [14]

- Houston included multifocal/G3 explants and TESLA intentionally enrolled extensive burden; therefore G1–2/unifocal restrictions are local conservatism, not universal trial criteria. Conversely, small mixed-tumor success does not rescue beyond-UCSF outcomes or validate advanced mixed inclusion. Negative limited biopsy cannot eliminate heterogeneous high-grade/mixed components.

Certainty / applicability: Low for small retrospective cohorts; very low for benefit and optimal thresholds in advanced disease. No randomized advanced-iCCA LT comparison or validated universal diameter/lesion-count limit.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- GUIDANCE: current NLRB small-cirrhotic unresectable solitary iCCA/mixed ≤ 3 cm after 6 months stability is a separate reviewed-exception lane; AASLD candidate wording uses < 3 cm. STUDY: SYS-TARE/Rennes use restricted burden; Houston/TESLA use broader burden. LOCAL: G1–2 and unifocal±adjacent satellites is a conservative first-study definition, not an evidence-proven optimum.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
SYS-TARE 2024 [6] "a single measurable lesion with or without associated peritumoral satellites" Published PDF pp2–3, Patients and Methods Source | Full-text PDF	A narrower studied morphology; it does not prove a safe upper diameter.
ILTS–ILCA consensus [9] "no agreement was reached regarding LT in this population." Consensus manuscript PDF p14, iCCA discussion Source | Full-text PDF	Refers to large/locally advanced or noncirrhotic iCCA; consensus did not endorse universal expansion.
AASLD/AST candidate evaluation [10] "a maximum diameter < 3 cm" AASLD/AST candidate-evaluation manuscript PDF p8, statement 11 Source | Full-text PDF	The clinical recommendation uses a strict inequality. Its rationale discusses solitary disease in cirrhosis; this is not the advanced-study population.
NLRB oncology guidance [2] "less than or equal to 3 cm" December 2025 oncology guidance p8, iCCA initial/extension criteria Source | Full-text PDF	Allocation guidance includes equality. Exactly 3.0 cm requires clinical adjudication; preserve the exact allocation wording instead of silently harmonizing it.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Keep the small-cirrhotic pathway separate; define the initial advanced study by pure, favorable-grade, localized morphology rather than an invented size cap. [6] [9] [10] [2] [3] [11] [12] [4] [13] [14]

- Proposed advanced entry phenotype: pure iCCA, G1–2, one mass-forming focus with or without adjacent peritumoral satellites; no separate multifocal deposits across distinct segments.

  - No universal 8 cm/three-lesion rule is adopted.

- Exclude known G3/poor differentiation, mixed HCC-iCCA, macroscopic tumor thrombus, proven malignant nodes, peritoneal or other extrahepatic disease from the initial advanced pilot.
  - Negative core sampling cannot guarantee the whole tumor grade; record this limitation.
- Intrahepatic vascular encasement without tumor thrombus may be considered if complete explant/reconstruction is technically feasible.
  - Larger size needs individualized technical/biological review, not an automatic permission.
- Do not apply the advanced-study mixed-histology exclusion to the separate NLRB small-cirrhotic route.
  - Preserve ≤3 cm guidance wording and flag the exactly-3.0 cm clinical discrepancy with <3 cm candidate-evaluation prose.

Why retain or change it: OPEN: initial advanced study accepts biopsy-confirmed pure iCCA, G1–2 on expert review, one dominant measurable mass with only adjacent peritumoral satellites, not separate multisegment/diffuse deposits. Do not invent 8 cm or 3 lesion limits. Exclude known G3, mixed histology, macroscopic tumor thrombus, proven nodal, peritoneal or extrahepatic disease from this first arm; allow intrahepatic vessel encasement without thrombus only if resection and transplant technical reviews agree. Record sampled grade uncertainty and all lesions centrally. Create separately approved expansion cohorts if G3, multifocal or mixed disease is to be studied.

INFORMATION REQUIRED BEFORE THE DECISION

Expert core-pathology review including pure versus mixed and grade; MRI/CT segment map; chest/systemic staging; nodal resolution; vascular thrombus versus encasement; resection review.

Clinical versus allocation: Advanced noncirrhotic or larger disease does not inherit the small-cirrhotic favorable NLRB pathway. Organ-route and research decisions remain explicit.

Roadmap / protocol reconciliation: Keep molecular profiling and conversion treatment from the iCCA roadmap; an actionable alteration is a treatment option, not a transplant biomarker.

Meeting record: iCCA-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

3.2 FINDINGS THAT HOLD OR STOP THIS PATHWAY

- Hold for uncertain primary or mixed histology, indeterminate nodes, unresolved distant lesions, unexplained marker rise or incomplete resectability review. Repeat imaging or targeted tissue assessment when it can resolve the decision.
- Active extrahepatic disease, malignant nodal/peritoneal staging or medical risk incompatible with LT stops this proposed liver-confined study route. Liver-only progression is handled by the OPEN iCCA-B2 hold/salvage rule. The initial iCCA-B1 proposal excludes G3/poor differentiation, mixed histology and macroscopic tumor thrombus; a broader arm is not activated.

4 PRETREATMENT WORKUP

- Order liver-protocol CT/MRI and chest imaging; obtain CBC, liver chemistry, bilirubin, albumin, INR, creatinine, CA19-9 and CEA, with infection/obstruction and nonsecretor context. Complete standard transplant fitness assessment if a clinically justified transplant evaluation is proceeding. [10]

- Use PET/CT and directed nodal assessment when appropriate for metastatic exclusion; a negative study cannot certify absence of microscopic disease. A mandatory repeated PET schedule is not a validated stand-alone transplant gate.
- Obtain molecular profiling appropriate for iCCA treatment choices and trial access. Record any assay/biopsy limitations. ctDNA may be collected under a defined correlative protocol but is not a required negative test for listing.
- Have HPB surgery, oncology and IR document resection, ablation, HAI and other liver-directed options. Before HAI or arterial therapy, review hepatic arterial anatomy and possible future reconstruction with the transplant surgeon. Do not imply that all HAI placement makes LT infeasible.

5 NEOADJUVANT THERAPY AND REASSESSMENT

5.1 SELECT A DISEASE TREATMENT PLAN WITHOUT ASSUMING TRANSPLANT EFFICACY

- Choose systemic treatment through the biliary oncology MDT and the corresponding cancer roadmap. Record regimen, dose intensity, toxicity, planned restaging and alternatives. Therapy is intended to control disease and test biology; transplant is not promised if imaging improves.
- Do not make gemcitabine/cisplatin/nab-paclitaxel the preferred intensification regimen. SWOG S1815 did not show a significant OS improvement over gemcitabine/cisplatin and added toxicity in advanced biliary cancer. [15]
- The GOLP randomized trial enrolled resectable high-risk iCCA. Conversion studies such as TransGOLP address a different population; neither establishes LT efficacy or post-LT ICI safety. Do not copy an 80% response claim or a transplant eligibility rule from these studies. [16] [17]
- Consider HAI or other liver-directed therapy according to expertise, anatomy and treatment goals, with explicit conversion-to-resection review. No Y90 dose, radiation schedule or modality is mandatory as a universal LT bridge. [18] [6]

5.2 DEFINE RESPONSE AND TIME UNDER OBSERVATION

- OPEN proposed advanced-disease rule: at least six consecutive months of response/stability on the effective treatment strategy after the most recent confirmed progression. Toxicity-driven changes with continuous control need not reset the clock; one liver-only salvage re-entry may be proposed under iCCA-B2. This is not proof that six months guarantees favorable biology. [5] [4]
- For the small-cirrhotic exception route, apply the current NLRB six-month stability wording and required lesion/no-new-disease conditions; document diagnosis, every treatment date and the allocation coordinator's interval determination. Do not silently substitute the advanced-study clock. [2]
- Proposed local schedule: contrast CT/MRI approximately every eight weeks during active therapy, with chest restaging and CA19-9 about every four weeks. Use RECIST 1.1 with original site mapping; mRECIST is not a validated iCCA LT eligibility system. Review earlier for symptoms, biochemical concern or toxicity.
- If ICI was given, record agent/last dose, immune toxicity and treatment, proposed interval, residual rejection risk and a graft-safety plan. An approximately 90-day planning safeguard is not a proven CCA-specific safe cutoff. Do not infer safety from a small selected no-rejection series. [19] [20] [21]

ICCA-B2 — WHAT DOES SUSTAINED CONTROL HAVE TO PROVE BEFORE TRANSPLANT ACTIVATION?

QUESTION AND COMPARISON

What disease-control duration, imaging definition and salvage re-entry rule should govern listing or activation for advanced iCCA while preserving conversion-to-resection opportunities?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Lunsford 2018 · Houston [5] Prospective case series: 21 referrals, 12 accepted and six LT recipients. Primary source | Full-text PDF Locator: Lancet Gastroenterol Hepatol 2018; 3: 337–348; Eligibility/Protocol, Table 1 and Figure 2; full local PDF. Review depth: Local full PDF reviewed	**Reported result:** Median diagnosis-to-LT interval 26 months. Post-LT 1-/3-/5-year OS 100%/83.3%/83.3%; five-year CI 27.3–97.5. Three of six recurred, median 7.6 months; RFS 50%. **Do not infer:** Six highly selected recipients, wide intervals and substantial pre-LT survival. Does not establish six months as optimal or one salvage line as the maximum.
McMillan 2022 · Houston update [4] Expanded Houston cohort, overlapping Lunsford. Primary source | TMC: retrieve PDF Locator: AJT 2022; Methods eligibility and §§3.1–3.3. Review depth: Publisher full HTML methods, patient flow, tables and outcomes inspected; complete PDF not found locally.	**Reported result:** Five listed patients converted to resection; seven others delisted and seven remained listed at cutoff. **Do not infer:** No timing or salvage comparator; keep referral, listing and LT clocks distinct.
SYS-TARE 2024 [6] Thirteen entrants; four LT recipients. Primary source | Full-text PDF Locator: Transpl Int 2024; Protocol, restaging and individual-recipient tables; full PDF. Review depth: Full text reviewed	**Reported result:** All four recipients had RECIST stable disease and Choi response, with median density reduction 52.5%. One FGFR 2-targeted pemigatinib rescue reached LT. Two of six staged patients had positive nodes. **Do not infer:** One rescue does not validate a general reset rule. PET-negative imaging cannot replace pathological staging.

Study • population • source	Results • interpretation boundary
Yaqub 2025 [13] Early prospective report; five LT recipients. Primary source | TMC: retrieve PDF Locator: BJS 2025; Figure 1 flowchart and Table 1. Review depth: Complete indexed primary research-letter text and Table 1/Figure 1 read; supplementary eligibility file not independently reviewed; PDF download failed.	**Reported result:** Two extensively multifocal patients recurred despite prolonged selection. **Do not infer:** Supplementary eligibility needs checking; time alone does not neutralize burden.
Giguet 2026 [14] Six-recipient retrospective series. Primary source | TMC: retrieve PDF Locator: JHEP Rep 2026; neoadjuvant/listing and Results sections. Review depth: Indexed complete publisher full text, Tables 1–3, cohort flow and late complications reviewed; downloadable PDF blocked. Six transplanted, not eight.	**Reported result:** Nineteen controlled younger patients underwent secondary resection. **Do not infer:** Observed longer selection does not prove a twelve-month requirement.
Schleimer 2026 · non-LT comparator [8] Two-center retrospective: 1, 407 total iCCA, 327 advanced liver-confined; 180 progressed/died before six months, 47 converted to resection, 39 potentially LT-eligible. Primary source | Full-text PDF Locator: Ann Surg Oncol 2026; survival methods, Results and Table 2; full PDF. Review depth: Original full PDF methods/results/Table 2 checked; diagnosis and six-month-landmark clocks distinguished	**Reported result:** Median survival 39 months from diagnosis; 3-/5-year survival 46%/26% from the six-month estimated-eligibility landmark. **Do not infer:** Favorable biology selects survivors before any LT. Historical 2008–2018 cohort, not a randomized treatment comparison.

Study • population • source	Results • interpretation boundary
PUMP-2 [18] Prospective phase II: 50 pump placements, 48 received FUDR, 37 concurrent GemCis. Primary source | Full-text PDF Locator: JCO 2026; PUMP-2 Methods/Results; full PDF. Review depth: Complete local publisher PDF inspected; methods, results and toxicity extracted; V6 abstract-only label corrected.	**Reported result:** Partial response 44%, disease control at six months 84%; five resected, one pathological complete response. Median survival 22.3 months, three-year 31.5%. **Do not infer:** Does not validate a pre-LT clearance test or subsequent arterial safety. Reassess resection after response.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

A six-month control requirement is a defensible tested entry rule, not an established biological truth. SYS-TARE, TESLA and Rennes used different selection intervals. Stable measurable disease can qualify without RECIST partial response. New lesions or extrahepatic spread cannot be dismissed as stable biology. Regression can change the correct operation to resection. [5] [4] [6] [13] [14] [8] [18]

- Successful later-line cases refute a categorical ban on all salvage but do not validate unrestricted repeated resets. Patients controlled for six months already have better non-LT prognosis. Comparing their post-LT survival with unselected advanced-cancer median survival exaggerates benefit. Unresolved lung, bone or nodal findings require adjudication before activation.

Certainty / applicability: Very low certainty for the optimal control clock, imaging metric or allowable salvage count; low certainty for sustained control as a useful selection approach.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- STUDY: six months in Houston/Rennes, four months in SYS-TARE and twelve months of treatment response in the TESLA flow diagram. LOCAL: approximately eight-week monitoring and one salvage re-entry. GUIDANCE: preserve the separate NLRB small-cirrhotic clock ('from diagnosis or last treatment'); do not silently replace it with the advanced research clock.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
Lunsford 2018 · Houston [5] “disease must have been controlled for 6 months on that regimen.” Published PDF p3, Methods—Study design and participants Source | Full-text PDF	Six months on an effective strategy has direct study precedent, including second-line treatment.

Reference and exact excerpt	What it supports / does not establish
McMillan 2022 · Houston update [4] "Patients who developed extrahepatic disease were delisted." Full text, section 2.2 Source | Local PDF pending	Liver-only progression was deferred and could be reconsidered; it was not treated identically to extrahepatic spread.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Require six months of continuous disease control, but distinguish liver-only progression with salvage from permanent exclusion. [5] [4] [6] [13] [14] [8] [18]

- Proposed clock: ≥6 consecutive months of radiological response/stability from initiation of the effective strategy after the most recent confirmed progression, with baseline imaging, no extrahepatic disease and persistent unresectability.
  - Image approximately every eight weeks and restage at completion.
- Dose reduction, toxicity-driven substitution or planned consolidation does not automatically reset the clock when control is continuous.
  - Confirmed liver-only progression pauses eligibility; allow one separately reported salvage re-entry after a fresh six months of control.
- Further progression after that salvage strategy, proven malignant nodes or extrahepatic spread excludes the patient from the initial study.
  - Reassess resection after each major response and immediately before LT; conversion to feasible curative resection redirects to resection.

Why retain or change it: OPEN: require at least six consecutive months without progression on the effective current strategy before activation. Record treatment start and first documented control after the latest progression. Obtain liver CT/MRI and chest staging approximately every eight weeks, with repeat resectability review. Accept RECIST stable disease or partial response with no new lesions or extrahepatic spread and a supportive biomarker trajectory; do not use CA19-9 alone. Dose reduction, toxicity-driven switching or planned consolidation need not restart the clock when control remains continuous. Permit one separately recorded salvage re-entry after a fresh six months of control and complete restaging. Further progression after that rescue, or proven extrahepatic/nodal disease, closes the initial arm. The one-rescue ceiling is local governance, not a validated published maximum.

INFORMATION REQUIRED BEFORE THE DECISION

Dated regimen/response timeline; comparable liver imaging and systemic staging; RECIST and multidisciplinary adjudication; CA19-9 in obstruction context; repeated HPB review. Record latest confirmed progression, current effective-strategy start, first control, each imaging assessment and the proposed activation date; repeat R0/functional-FLR review after meaningful response.

Clinical versus allocation: This is the advanced-study clock; do not substitute it for the separate small-cirrhotic NLRB observation/exception requirements.

Roadmap / protocol reconciliation: Preserve systemic/HAI/liver-directed conversion options. The V5 blanket progression-stop wording is replaced by hold, approved salvage re-entry and definitive study exclusion.

Meeting record: iCCA-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

6 RESPONSE ASSESSMENT AND DECISION TO ADVANCE

- Advance to final staging only when serial imaging shows response/stability without new disease, liver/medical reserve remains acceptable, toxicity has recovered, and independent review still finds no durable resection/ablation option. Document what changed since the previous surgical assessment.
- If conversion to complete resection becomes feasible, reassess that route before LT; record conversion as a treatment success, not transplant-program failure. A transplanting study must not keep a now-resectable patient merely to preserve its recipient denominator. [4] [8]
- Hold for uncertain nodal/extrahepatic findings or marker-imaging discordance. Confirmed liver-only progression pauses eligibility; the OPEN iCCA-B2 proposal permits one separately reported salvage re-entry after a fresh six months of control. Further progression after salvage, new extrahepatic disease or malignant operative staging excludes the initial study. Do not escalate toxic therapy solely to manufacture eligibility.

7 LISTING AND ORGAN ALLOCATION

- Advanced iCCA is not a standardized exception. A nonstandard narrative may be submitted, but approval and score are uncertain. Meeting a study rule or possessing a donor does not create exception eligibility. Record calculated MELD separately. [1] [2]
- Before recipient activation, document the intended DDLT or LDLT route, actual exception decision if applicable, expected waiting time, and a recipient-benefit assessment against contemporary non-LT care. Do not promise priority or treat DCD, an extended-criteria graft or machine perfusion as evidence that no other candidate could benefit. [1] [10]
- For LDLT, retain independent donor advocacy and donor-team assessment, freedom to withdraw, and separate consent covering uncertain recipient benefit, operative and long-term morbidity, recovery, financial burden and possible retransplantation. Do not relax donor safety criteria or make possession of a living donor a clinical eligibility requirement. [1] [10] [22]
- The living-donor literature reports low but nonzero mortality and incomplete recovery follow-up. Use current center- and graft-specific counseling; do not pool overlapping registry studies or portray donor survival as proof of complete recovery. [22]

ALLOC-B1 — WHAT ARE WE APPROVING?

Evidence and uncertainty before the shared rule

The separation is a governance requirement, not an empirical survival cutoff. A clinically appropriate patient can lack exception eligibility, and an approved exception does not prove clinical benefit. LDLT requires an independent donor determination; a registry entry is not permission to perform experimental transplantation. [1] [2] [23] [24]

Evidence-to-threshold distinction: Six domains: local checklist architecture, not six mandatory national approvals.; 100% applicable completion: governance safeguard, not an outcome-derived percentage.; Hold

pending required authorization: operational implementation of separate clinical, donor and research responsibilities.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Require 100% completion of the applicable clinical, allocation, donor and research determinations before activation. [2] [1] [23] [24]

- Complete six separate dated determinations: clinical suitability; standardized-exception eligibility; nonstandard NLRB review status; calculated-MELD access; LDLT donor clearance; and research authorization.
  - Mark nonapplicable domains explicitly with a reason—do not manufacture six approvals for every patient.
- Any required missing, pending or adverse determination is a HOLD.
  - An exception application is not approval, a donor is not recipient benefit, and clinical approval is not study activation.
- Recheck the current policy/guidance version before each request; record actual decision and priority rather than predicted entitlement.

Why retain or change it: Retain 100% completion of applicable determinations with dated responsible owners and reasoned N/A entries. A required unresolved determination means HOLD, not an assumed refusal or an authorization by default.

INFORMATION REQUIRED BEFORE THE DECISION

Signed suitability record; exact allocation route and policy version; exception outcome; donor-team status; study and institutional activation record.

Clinical versus allocation: Policy 9.4 permits discretionary requests where appropriate; absence of favorable diagnosis-specific guidance is not absence of a review mechanism, and a request does not promise approval.

Roadmap / protocol reconciliation: Use the same clinical-versus-allocation separation in every standard and high-risk pathway.

Meeting record: ALLOC-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

ALLOC-B2 — WHAT IS THE RELEVANT ORGAN OPPORTUNITY COST?

Evidence and uncertainty before the shared rule

Use pathway-level benefit from an agreed entry date, not recipient-only survival. Compare with contemporary tumor-specific care and actual local access. LDLT avoids one direct deceased-donor allocation but creates healthy-donor risk and may generate rescue graft use. Graft quality/perfusion changes do not make opportunity cost zero. [24] [1] [25] [22] [26] [27] [28]

Evidence-to-threshold distinction: Two reviewers: local independence safeguard, not validated optimum.; 6/12/18 months: sensitivity scenarios for access/dropout, not mandatory waits.; HCC ≥60% five-year ITT OS: one expert trial-design benchmark; not a cross-cancer allocation policy.; All grafts: count initial and rescue/retransplant grafts; donor morbidity remains a separate outcome.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Require two independent recipient-benefit reviews and an explicit organ-use case; do not adopt a universal five-year survival cutoff. [27] [22] [24] [1] [25] [26] [28]

- Proposed gate: two reviewers not both responsible for recruitment must document expected incremental benefit over the actual contemporary alternative, with unresolved disagreement referred to the full committee.
- For DDLT, show local expected access/dropout and sensitivity scenarios at 6, 12 and 18 months, total graft use including retransplantation, and plausible alternative recipients.
  - These are planning scenarios, not imposed waiting periods.
- For LDLT, separately show healthy-donor risk and potential downstream deceased-donor graft use.
  - Do not call DCD, extended-criteria grafts or machine perfusion "zero opportunity cost." No priority changes follow from an AI-selected survival threshold.

Why retain or change it: Retain independent benefit and organ-use reviews, but require disease-specific ITT evidence and explicit uncertainty. No universal five-year survival cutoff or numerical allocation weight is adopted.

INFORMATION REQUIRED BEFORE THE DECISION

Same-eligibility/time-zero comparator; expected access; dropout; donor/graft type; uncertainty intervals; graft loss/retransplant estimate; written independent reviews.

Clinical versus allocation: Only the applicable allocation process determines priority. Favorable post-LT prognosis, evidence grade and urgency are not interchangeable measures.

Roadmap / protocol reconciliation: Compare against the cancer roadmap's actual resection, ablation, HAI, radiation or systemic alternative—not historical untreated survival.

Meeting record: ALLOC-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

ALLOC-B3 — WHAT MUST DONOR CONSENT AND FOLLOW-UP INCLUDE?

Evidence and uncertainty before the shared rule

Donor mortality is low but not zero; complication, recovery and financial burdens are more frequent. Registry missingness and selected healthy donors limit reassurance. Consent must include uncertain recipient benefit, recurrence and possible early graft loss, without pressure to proceed or relax donor criteria. [22] [29] [30] [31] [1]

Quantitative donor context: Kelly reports 8/8,619 deaths by 90 days across eras; A2ALL recorded ≥1 complication in 296/740 donors. These are different outcomes/eras, not poolable rates. The complete donor evidence and follow-up limitations are in ALLOC-B3.

Evidence-to-threshold distinction: Baseline/6 weeks/6 months/1/2 years then annually to five: proposed support schedule; current mandatory reporting remains separately checked.; One sentinel event: conservative review/pause trigger, not a statistically estimated harm boundary.; No relaxed donor criteria: protection against shifting uncertain recipient benefit onto a healthy person.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Do not relax donor criteria; fund independent follow-up and pause after a serious sentinel donor event. [1] [22] [29] [30] [31]

- Proposed readiness gate: independent donor evaluation/advocacy, documented freedom to withdraw, recipient-benefit uncertainty consent, and funded medical, psychological, recovery and financial follow-up before investigational LDLT.
- Proposed local follow-up: baseline, six weeks, six months, one year, two years, then annually through five years.
  - Meet current OPTN reporting duties separately; this five-year support plan is not claimed to be a national reporting requirement.
- One donor death, life-threatening complication, permanent major disability or liver failure requiring transplantation triggers an immediate pause of the relevant investigational LDLT activity and independent review.
  - Other serious donor harm also warrants review; reopening needs independent authorization.

Why retain or change it: Retain independent advocacy, ability to withdraw, funded medical/psychological/financial follow-up and sentinel-event pause. Add explicit denominator/era caveats to consent estimates and collect local outcomes prospectively.

INFORMATION REQUIRED BEFORE THE DECISION

Graft/resection-specific and center-specific risk; donor recovery/employment baseline; independent consent and advocate records; funded follow-up/support; sentinel-event response owner.

Clinical versus allocation: LDLT clearance is separate from recipient selection; the potential need for retransplantation must be included in organ-use discussions.

Roadmap / protocol reconciliation: The same independent donor safeguards apply whether recipient disease is standard or experimental.

Meeting record: ALLOC-B3 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

ICCA-B3 — DOES ACCESS TO LDLT CHANGE THE EVIDENTIARY THRESHOLD?

QUESTION AND COMPARISON

Does current recipient-benefit and donor-risk evidence justify an advanced-iCCA LDLT study, and what must be in place before an individual donor/recipient pair can proceed?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Sethi 2026 · iCCA LDLT [7] Retrospective single-center seven-recipient LDLT series. Primary source | TMC: retrieve PDF Locator: AJT online 29 September 2026; primary abstract, PMID 42 810 588. Review depth: Primary abstract independently verified in this pass; prior external full-text review is not treated as this lane's complete-PDF verification.	**Reported result:** Median follow-up 5.1 years; two recurrences, one death. Five-year OS 85.7% (CI 33.4–97.9), RFS 71.4% (CI 25.8–92.0). Two gradeIII complications within six months; all donors alive. **Do not infer:** Abstract independently checked; full selection, screened denominator and donor morbidity require PDF. Seven recipients cannot establish net benefit.
Lunsford 2018 · Houston [5] Twenty-one referrals, twelve accepted, six recipients. Primary source | Full-text PDF Locator: Lancet Gastroenterol Hepatol 2018; Methods/Results; local PDF. Review depth: Local full PDF reviewed	**Reported result:** Five-year post-LT survival 83.3% with very wide CI; three of six recurred. **Do not infer:** Feasibility, not LDLT-specific causal benefit; overlaps McMillan.
Giguet 2026 [14] Six selected deceased-donor recipients. Primary source | TMC: retrieve PDF Locator: JHEP Rep 2026; Table 3 and late-complication Results. Review depth: Indexed complete publisher full text, Tables 1–3, cohort flow and late complications reviewed; downloadable PDF blocked. Six transplanted, not eight.	**Reported result:** Arterial complications 3/6 and biliary 4/6; one late retransplant and one septic-cholangitis death at 6.4 years despite five-year OS 100%. **Do not infer:** High survival is not low morbidity or cure; SIRT, surgery and graft-quality effects cannot be separated.
Schleimer 2026 · non-LT comparator [8] Thirty-nine selected non-LT patients from 327 advanced liver-confined cases; 22 received HAIP. Primary source | Full-text PDF Locator: Ann Surg Oncol 2026; Methods/Results/Table 2; local PDF. Review depth: Original full PDF methods/results/Table 2 checked; diagnosis and six-month-landmark clocks distinguished	**Reported result:** Median 39 months from diagnosis; five-year survival 26% from the six-month landmark. **Do not infer:** More relevant than unselected metastatic BTC but historical and nonrandomized; cannot yield precise individual LT benefit.

Study • population • source	Results • interpretation boundary
PUMP-2 [18] Fifty pump placements; 48 received FUDR; single-arm phaseII. Primary source | Full-text PDF Locator: JCO 2026 PUMP-2; full-PDF Methods/Results and adverse events. Review depth: Complete local publisher PDF inspected; methods, results and toxicity extracted; V6 abstract-only label corrected.	**Reported result:** Median survival 22.3 months; three-year 31.5%; five conversion resections. Arterial dissection prevented treatment in one patient. **Do not infer:** A current regional option, not a randomized LDLT comparator or proof of later graft-arterial safety.
Oh 2025 [32] Randomized phaseIII advanced BTC: 685 participants, 341 durvalumab/GemCis and 344 placebo/GemCis. Primary source | TMC: retrieve PDF Locator: J Hepatol 2025; 83: 1092–1101; primary three-year update. Review depth: Primary article abstract/results and trial design verified; complete PDF not reviewed in this lane.	**Reported result:** Median OS 12.9 versus 11.3 months; HR 0.74 (CI 0.63–0.87), 36-month OS 14.6% versus 6.9%, median follow-up 41.3 months. **Do not infer:** High-level evidence for non-LT therapy, not pre-LT ICI safety or an exchangeable control for LT recipients.
Kelly 2026 [22] SRTR 1989–2023: 8, 981 eligible donors, 8, 619 mortality analysis and 8, 314 secondary follow-up. Primary source | Full-text PDF Locator: J Transplant 2026: 2 545 887; Missing Data and Results§§3.1–3.3; complete 8-page PDF. Review depth: Original full PDF checked	**Reported result:** Eight 90-day deaths (0.1%), 18 within two years (0.2%). Contemporary 90-day mortality approximately 0.04%. Secondary-outcome completeness 60–67% at six months and 37–39% at two years. **Do not infer:** All-cause deaths are not all donation-attributable. Missingness, historical eras and mixed graft types limit individual risk estimates; employment burden also matters.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

The seven-recipient LDLT experience supports developing a carefully governed study, not announcing a standard indication. A shorter wait may avoid dropout but can also remove the biological observation that selected successful cohorts. Donor availability does not erase recipient uncertainty or non-LT alternatives. Benefit and donor harm require separate analyses and independent consent. [7] [5] [14] [8] [18] [32] [22]

- No randomized advanced-iCCA LDLT-benefit trial or validated minimum five-year survival forecast establishes study readiness. Response-selected non-LT patients may live for years. National mortality is small but nonzero and does not capture all financial, psychological or functional burden. Seven living donors reported alive is not a safety guarantee.

Certainty / applicability: Very low certainty for causal LDLT benefit and the disease-specific donor-risk tradeoff; moderate confidence in broad national donor-mortality estimates, with limited individual transferability.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- POLICY: donor independence, informed consent and withdrawal protections remain mandatory. STUDY: seven-recipient LDLT feasibility with wide confidence intervals; no validated minimum survival forecast. LOCAL: six-domain readiness gate. Recipient boundaries and safety stopping rules require prospective institutional approval before the first pair.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
Sethi 2026 · iCCA LDLT [7] "This retrospective, single-center study investigated outcomes for seven patients" 2026 published-paper abstract; complete local PDF pending Source | Local PDF pending	Justifies careful prospective development, not routine LDLT access. This pass independently verified the abstract, not the complete methods.
Schleimer 2026 · non-LT comparator [8] "Their median survival from diagnosis was 39 months" Published PDF p1 / journal p6099, abstract Results Source | Full-text PDF	Selected contemporary non-LT survival is a relevant comparator; this is not a randomized LT-benefit estimate.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Develop a prospective LDLT feasibility study, but do not permit transplantation under this proposal yet. [7] [8] [5] [14] [18] [32] [22]

- Activation cutoff: all six readiness domains complete—approved recipient rules; independent benefit/comparator review; separate donor/advocate clearance; explicit uncertainty/graft-use consent; funded donor follow-up; and an all-screened registry with independently approved safety monitoring.
- Donor availability never lowers iCCA-B1/B2 selection or control requirements.
  - Retain ordinary donor safety criteria and independent withdrawal rights.
- Do not use predicted five-year survival ≥50%, or another unvalidated number, as proof of readiness.
  - Until approvals are complete: screening/registry and study development only, no protocol-directed LT.

Why retain or change it: OPEN: develop the protocol, but do not activate clinical transplantation yet. Require six readiness domains: fixed B 1/B 2 recipient rules and independent HPB review; documented best non-LT comparator and individualized expected benefit; independent donor advocacy, medical and psychosocial clearance with center-specific risks and withdrawal protection; explicit recipient/donor

uncertainty consent; funded long-term follow-up; and an all-screened registry with prespecified safety review and independent IRB/DSMB governance. A willing donor must not relax tumor-burden or control requirements, or enable an unapproved ICI/regimen arm.

INFORMATION REQUIRED BEFORE THE DECISION

Written study, statistical/safety plan, non-LT alternatives including HAI/conversion, donor resources, consent forms and documented independent approvals.

Clinical versus allocation: LDLT avoids allocation of that deceased-donor graft but adds healthy-donor risk and potential retransplant demand; it does not itself prove net benefit.

Roadmap / protocol reconciliation: Continue the iCCA treatment roadmap while developing the study; document which contemporary alternative the candidate would otherwise receive.

Meeting record: iCCA-B3 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

8 OPERATIVE STAGING AND SURGERY

- Define staging laparoscopy/laparotomy and appropriate nodal/peritoneal assessment in the approved study before transplant. Sample suspicious findings; use frozen section where it affects the operation. Do not proceed with unresolved malignant findings or treat a negative PET as a substitute. [5] [4]
- Complete native hepatectomy with oncologic handling and planned vascular/biliary reconstruction if LT is authorized. An additional “extended hepatectomy” is not a separate maneuver during total native hepatectomy. Unexpected metastatic disease stops this study operation.
- Explant pathology must map viable disease, differentiation, mixed components, margins, vascular/nodal findings and treatment response for the prespecified analytic stratum.

9 POST TRANSPLANT IMMUNOSUPPRESSION

- Use institutional induction and maintenance with pharmacy review of graft, renal, infection and oncologic risks. No iCCA-specific regimen has proven recurrence benefit. Do not mandate early steroid withdrawal or empiric high tacrolimus because of prior ICI. [10] [19]
- If considering an mTOR inhibitor, follow its current label and defer initiation until appropriate healing and vascular stability. Antitumor properties are not proof of clinical prevention of iCCA recurrence.
- Monitor graft function closely and investigate unexplained abnormalities, including infection, vascular/biliary causes and rejection. For-cause biopsy is appropriate when needed; fixed protocol biopsies require their own justified research plan.

10 ADJUVANT THERAPY

- There is no established routine post-LT adjuvant regimen for advanced iCCA. Residual viable tumor does not automatically mandate capecitabine, gemcitabine or targeted therapy. An actionable FGFR/IDH alteration without recurrence is not an adjuvant indication.
- Any postoperative systemic study must separately define recovery requirements, dose/interaction evaluation, infection and marrow monitoring, and graft-related stopping rules. Pre-LT drug activity is not a post-LT safety result.

11 SURVEILLANCE

- Proposed local schedule for approval: chest and contrast abdominal/pelvic imaging with CA19-9 and CEA every three months for two years, every six months through year five, then individualized follow-up. Choose CT/MRI according to clinical circumstances. This is not a trial-proven optimal schedule.
- Use PET or biopsy to resolve a specific concern. ctDNA is exploratory and should trigger confirmatory assessment when discordant, not automatic treatment or delisting.

12 RECURRENCE MANAGEMENT

- Restage and review suspected recurrence with the biliary and transplant MDT. Local treatment may suit selected limited disease; systemic options depend on molecular findings, graft function, interactions and patient goals. Do not prescribe uniformly aggressive multimodality treatment.
- Individualize immunosuppression adjustment. Post-LT checkpoint therapy is not routine and carries rejection/graft-loss risk distinct from pre-LT exposure. [33]

13 OUTCOMES AND SAFETY MONITORING

- Register every referred or screened candidate in the institutionally approved system, not only recipients. Record acceptance or rejection, treatment entry, response, listing, exception decision, conversion to resection or ablation, progression, toxicity, decompensation, death, patient withdrawal and access-related loss.
- Report overall survival from the prespecified entry point and separately from listing and transplantation. Keep recurrence-free survival, recurrence incidence, graft survival and survival after recurrence distinct, with definitions, confidence intervals, numbers at risk and follow-up completeness. Do not call recipient-only survival an intention-to-treat outcome.
- Capture 90-day recipient morbidity and mortality, biopsy-defined rejection, infection, arterial and biliary complications, graft loss, retransplantation and total grafts used. Record recurrence site, salvage treatment, time off systemic therapy, functional recovery and quality of life, not survival alone.
- Before a transplanting study opens, biostatistics and the independent safety board must approve indication-specific monitoring, evaluable denominators, pause authority and reopening criteria. ALLOC-B4 now proposes explicit initial-cohort review triggers for ratification; they are not validated efficacy/futility boundaries. No universal cross-cancer five-year survival cutoff is adopted.
- Use a contemporary comparator with the same eligibility and time zero. Account for the interval needed to survive to transplantation, crossover, competing risks and overlapping study cohorts. A nonsignificant difference is not proof of equivalence.

ALLOC-B4 — WHAT DATA ARE REQUIRED BEFORE EXPANSION?

Evidence and uncertainty before the shared rule

A small pilot can test feasibility, staging fidelity, data capture and acute safety, not prove durable survival benefit. Define time zero, all-screened enrollment, waitlist failures, post-LT survival/recurrence, competing death, all grafts, donor harms and patient-reported outcomes before opening. An independent board needs actual pause/reopening authority. [25] [34] [23] [22] [6] [13]

Zero observed events among ten fully followed independent patients still has a one-sided exact 95% binomial upper bound of 25.9% (1–0.05^(1/10)). Thus ten survivors cannot establish a <5% event risk. Ninety percent

data completeness permits one missing patient; safety events and vital status require active ascertainment, not tolerance of unaccounted loss.

Evidence-to-threshold distinction: Ten per stratum and interim after five at 90 days: local feasibility cap and safety review cadence.; Twelve-month minimum: earliest expansion discussion, not proof of long-term cancer benefit.; ≥90%: proposed non-vital key-measure completeness threshold; not permission to ignore deaths/graft events or unknown status.; One early death/graft loss/re-LT: review/pause trigger, not automatic futility or withholding rescue.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Use a small staged pilot with all-screened accounting and explicit safety pauses; no automatic expansion on favorable recipient survival. [25] [34] [23] [22] [6] [13]

- Proposed initial cap: ten transplants per separately approved biological stratum.
  - Review after the first five have 90-day follow-up; no expansion decision until all first ten have at least 12-month follow-up.
  - Reconcile vital status, graft events and known serious harms for every participant; require ≥90% completeness for other prespecified key outcomes.
- In an initial experimental cohort, one all-cause recipient death, graft loss or retransplantation within 90 days triggers a pause of further experimental enrollment/transplantation pending independent review.
  - A later possibly treatment-related death/graft loss or a donor sentinel event also triggers review/pause.
  - Rescue care is never withheld; reopening requires independent authorization.
- These are proposed conservative review triggers, not statistically validated efficacy/futility boundaries.
  - Before opening, biostatistics and the independent board must ratify or replace them with indication-specific rules, denominators, pause authority and reopening criteria.
- Capture every screened candidate, including nonrecipients, conversion to resection, dropout, deaths and access barriers; report survival from entry and from LT, recurrence, graft use, donor recovery and quality of life.
  - Ten survivors alone do not establish benefit or authorize expansion.

Why retain or change it: Retain the small staged pilot as an OPEN governance proposal, not an efficacy claim. Clarify that known safety/vital-status data must be reconciled for every enrolled patient; ≥90% applies only to other prespecified key measures. Indication-specific statistical design may replace the proposed numbers before opening.

INFORMATION REQUIRED BEFORE THE DECISION

Locked eligibility strata; prospective entry/time-zero dataset; modern comparator; adverse-event definitions; follow-up-completeness report; independently signed statistical and safety plan.

Clinical versus allocation: Study monitoring never changes exception criteria or guarantees future access. Audit access by referral, geography, resources and donor availability without turning protected characteristics into exclusion rules.

Roadmap / protocol reconciliation: Retain cancer-roadmap treatment alternatives and document why non-LT care was chosen when candidates do not proceed.

Meeting record: ALLOC-B4 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

14 MULTIDISCIPLINARY RESPONSIBILITY

- Transplant/HPB surgery, hepatology, oncology, IR, radiology and pathology own selection and treatment interpretation. Pharmacy owns drug interaction and perioperative review; the donor team is independent. Research methods and safety oversight require designated accountable reviewers.

15 DECISIONS REQUIRED BEFORE EXPANSION

Resolve iCCA-B1–B3 and ALLOC-B1–B4 before study activation. Record final wording, population, histology strata, comparator, response clock, organ route and safety authority. Pure iCCA and mixed HCC–iCCA outcomes must remain separately reportable.

Evidence review record: Five analyst comments and reconciliation | Prospective outcomes dataset

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[34] Eldridge, Sandra M; Chan, Claire L; Campbell, Michael J; et al. CONSORT 2010 statement: extension to randomised pilot and feasibility trials. BMJ. 2016;:i5239. DOI | Local PDF pending

Clinical research / guidance. Primary journal HTML checklist, scope and explanations reviewed; complete PDF pending.

REVISION NOTE

Revised 2026-10-24 06:55 CDT. Evidence-before-recommendation review of all 27 B questions. Earlier protocol versions and external analyst records are preserved; this is an Analyst A synthesis, not a new external-analyst vote.

Problem: The earlier B proposals had an uneven evidence base and emphasized a stance before its supporting and contrary data. **Correction and reason:** Added question-level primary-study tables with selection, denominators, outcome clocks, harms, limitations, exact source excerpts and threshold provenance before each OPEN proposal. Sources/PDF links are adjacent; missing full texts remain flagged.

Problem: Local operational numbers could be mistaken for validated eligibility or allocation thresholds. **Correction and reason:** Separated outcome-derived estimates, guideline/policy wording and local safeguards; revised proposals when the deeper review changed their evidentiary basis. Institutional decisions remain blank.

Approval record Clinical lead _____ Allocation review _____ Research authorization if required _____ Donor review if applicable _____ Date _____